

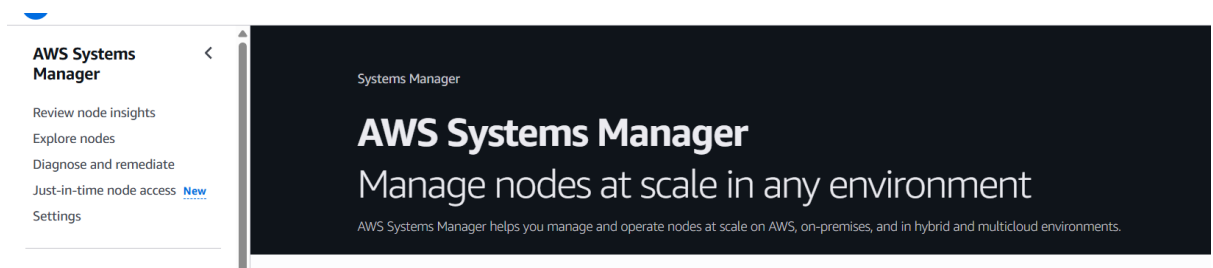
AWS Systems Manager Parameter Store

AWS Systems Manager Parameter Store provides secure, hierarchical storage for configuration data management and secrets management. You can store data such as database connection strings, application configuration values, license codes, and passwords as parameter values.

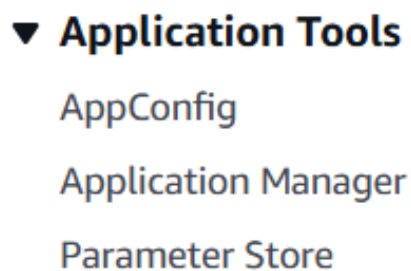
- Implemented secure secret management using AWS Systems Manager Parameter Store and IAM Roles, allowing EC2 instances to retrieve encrypted configuration values dynamically without hardcoding credentials.

Implementation Steps:

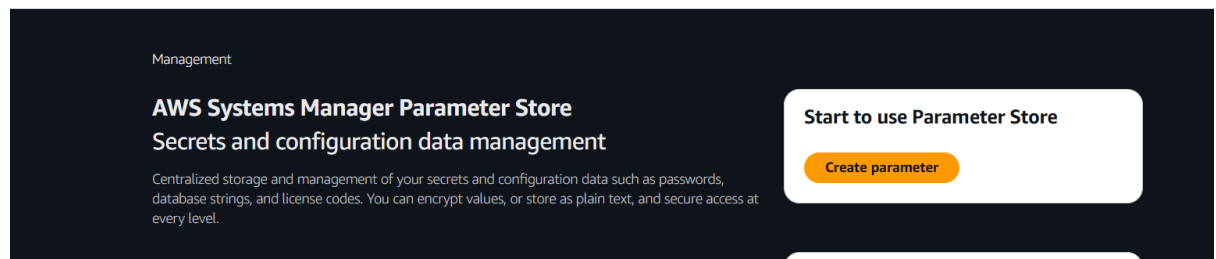
1. AWS Console and Search 'Systems Manager' Click Systems Manager



2. In the left panel we can see parameter store in Application Tools



3. Click on create parameter



4. parameter details, type and value

Create parameter

Parameter details

Name

/prod/db/password

When naming a parameter, you can use forward slashes (/) to organize it into a hierarchy. [Learn more about hierarchies](#)

Description — Optional

Type

- ☐ String
Any string value.
- ☐ StringList
Separate strings using commas.
- ☒ SecureString
Encrypt sensitive data using KMS keys from your account or another account.

KMS key source - [Learn more](#)

- ☒ My current account
Use the default KMS key for this account or specify a customer-managed key for this account.
- ☐ Another account
Use a KMS key from another account

KMS Key ID

alias/aws/ssm

You have selected the default AWS managed key
AWS managed keys cannot be shared with other AWS accounts, and all users in this AWS account and Region have access to the key. Use a customer managed key for sharing and access control.
[Learn more](#)

Value

.....

5. Created parameter

☰ AWS Systems Manager > Parameter Store

Create parameter request succeeded! [View details](#)

Parameter Store

[My parameters](#) | [Public parameters](#) | [Settings](#)

Ready to scale? Enable higher throughput to handle more API calls - [Manage Settings](#) to increase TPS. [Manage settings](#)

My parameters

[View details](#) [Edit](#) [Delete](#) [Create parameter](#)

Filter parameters

☐	Name	Tier	Type	Last modified
☐	/prod/db/password	Standard	SecureString	Fri, 08 May 2026 06:59:47 GMT

6. Created a role by selecting AWS Service and EC2

☰ IAM > Roles > Create role

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Select trusted entity [Info](#)

Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or other

Service or use case
EC2

7. Attached permissions

Add permissions [Info](#)

Permissions policies (1/1152) [Info](#)

Choose one or more policies to attach to your new role.

Filter by Type: [All types](#)

🔍 AmazonSSMReadOnlyAccess [✕](#)

☒	Policy name ↗	Type
☒	 AmazonSSMReadOnlyAccess	AWS managed

8. Given Role name and created SSM Role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+-._@-' characters.

Description

Add a short explanation for this role.

9. Created new Ec2 instance in that Advanced Details Attached EC2 SSM Role

☰

[EC2](#) > [Instances](#) > Launch an instance

▼

Advanced details [Info](#)

Domain join directory | [Info](#)

Select ▼

[Create new directory](#)

IAM instance profile | [Info](#)

EC2-SSM-Role
arn:aws:iam::928430096503:instance-profile/EC2-SSM-Role ▼

[Create new IAM profile](#)

Hostname type | [Info](#)

IP name ▼

10. After that Connect with AWS CLI

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ec2-user@ec2-13-221-116-172.compute-1.amazonaws.com
The authenticity of host 'ec2-13-221-116-172.compute-1.amazonaws.com (13.221.116.172)' can't be established.
ED25519 key fingerprint is SHA256:HLbR785cAH7X4MbXQCLTmqNjo7NSs+surIpjw8zfmJk.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-221-116-172.compute-1.amazonaws.com' (ED25519)
to the list of known hosts.
```

```
~_##### Amazon Linux 2023
~\#####\
~\###|
~\#/\
~~~~V~'-> https://aws.amazon.com/linux/amazon-linux-2023
~~~~/m/'
```

```
[ec2-user@ip-172-31-31-185 ~]$ aws --version
aws-cli/2.33.15 Python/3.9.25 Linux/6.1.168-203.330.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-172-31-31-185 ~]$ aws ssm get-parameter \
--name "/prod/db/password" \
--with-decryption
{
  "Parameter": {
    "Name": "/prod/db/password",
    "Type": "SecureString",
    "Value": "MySecret123",
    "Version": 1,
    "LastModifiedDate": "2026-05-08T06:59:47.027000+00:00",
    "ARN": "arn:aws:ssm:us-east-1:928430096503:parameter/prod/db/password",
    "DataType": "text"
  }
}
```

```
[ec2-user@ip-172-31-31-185 ~]$ |
```